COMMERCIAL PARTS LIST

ITEM	QTY	PART NUMBER	DESCRIPTION
C1	1	2000.23 IG 2500:1	HIGH VOLTAGE DIVIDER, 0.1% RATIO TOL, 0.25% ABS TOL, 15 PPM RATIO TCR, 30 PPM RATIO ABS TCR, EBG RESISTORS
C2	1	D38999/20FC35PN	WALL MOUNT RECEPTACLE, THREADED COUPLING, SHELL SIZE C, 22 SIZE 22D PINS, MIL-DTL-38999
C3	22	M39029/58-360	CONNECTOR CONTACT PIN, SIZE 22D, MIL-C-39029
C4	1	F404B9	HIGH VOLTAGE RECEPTACLE, 30KV DC, ALDEN PRODUCTS
C5	1	M85049/95-14A	NUT PLATE, 3/4 PERIMETER, FLANGE MOUNT RECEPTACLE, SHELL SIZE C
C6	2	MOX92021006FVE	HIGH VOLTAGE RESISTOR, SUPER MOX SERIES, OHMITE MFG.
C7	8	MS15795-803	FLAT WASHER, 300 SERIES SSTL, NASM15795, .125 ID X .250 OD X .022 THK
C8	4	MS21043-04	NUT, SELF LOCKING, .112-40 UNJC-3B, REDUCED HEX, REDUCED HEIGHT, CRES, FLANGED
C9	2	MU18-4R/SK	RING TONGUE, NON-INSULATED BUTTED SEAM, 22-18 AWG, 3M SCOTCHLOK
C10	3	MU18-8R/LK	RING TONGUE, NON-INSULATED BUTTED SEAM, 22-18 AWG, 3M SCOTCHLOK
C11	2	NA0068A040008	PAN HEAD SCREW, OFFSET CRUCIFORM, METRIC, A286 CRES, MJ4 X 0.7, 8 MM LONG
C12	12	NASI101E02-4	SCREW, MACHINE-FLAT FILLISTER HEAD .086-56 UNJC-3A X .250 LENGTH
C13	4	NASI101E04-6	SCREW, MACHINE-FLAT FILLISTER HEAD .112-40 UNJC-3A X .375 LENGTH
C14	4	NASI101E06-18	SCREW, MACHINE-FLAT FILLISTER HEAD .138-32 UNJC-3A X 1.125 LENGTH
C15	5	NASI102E06-5	.138-32 UNJC-3A X .313 FLAT 100 DEG HEAD MACHINE SCREW
C16	12	NASI149EN232R	NO. 2 FLAT WASHER .099" ID X .250" OD X .032" THK
C17	4	NASI352N04-10	#4-40(.112) UNRC-3A X .625 SOCKET HEAD CAP SCREW
C18	4	NAS620C4L	#4 REDUCED THICKNESS FLAT WASHER REDUCED O.D.
C19	4	NAS620C6L	#6 REDUCED THICKNESS FLAT WASHER REDUCED O.D.
C20	4	NTE517	SILICON HIGH VOLTAGE PLASTIC RECTIFIER, NTE ELECTRONICS
C21	A/R	PLTIM-M702Y	CABLE TIE, HALAR FLUOROPOLYMER, PANDUIT CORP.
C22	3	TC105	CABLE TIE MOUNTING BASE, ALUMINUM, THOMAS & BETTS
C23	A/R	2216 B/A	2216 B/A GRAY EPOXY 3M SCOTCH WELD EPOXY
C24	A/R	3145 CLEAR	RTV ADHESIVE/SEALANT, CLEAR, DOW CORNING
C25	A/R	M22759/11-22-0	22 AWG WIRE, SILVER COATED STRANDED COPPER, FLUOROPOLYMER INSULATED, EXTRUDED TFE, BLACK
C26	A/R	RTV566	SILICONE RUBBER COMPOUND, MOMENTIVE PERFORMANCE MATERIALS
C27	A/R	SN63PB37/RMA	SOLDER, TIN/LEAD, RMA FLUX
C28	A/R	W-SI-20KV-1000	ULTRAVOLT 20 KV WIRE

CRIMP TOOL LIST

CONTACT P/N	CRIMP TOOL	POSITIONER	INSERT/EXTRACT	REF. DES.
M39029/58-360	M22520/2-01	M22520/2-09	M81969/14-01	J9
MU18-4R/SK	TR-490 OR EQUIV.	N/A	N/A	N/A
MU18-8R/LK	TR-490 OR EQUIV.	N/A	N/A	N/A

TENSILE PULL STRENGTH LIST

MANUFACTURER	CONTACT P/N	WIRE AWG	LBS
QPL	M39029/58-360	22	13
3M	MU18-4R/SK	22	13
3M	MU18-8R/LK	22	13

16. FILL CAVITY OF ITEM 5 (PASSIVE CIRCUIT BASE) WITH ITEM C26 (SILICONE RUBBER) FLUSH TO SURFACE INDICATED AFTER ASSEMBLY AND TEST.

17. TORQUE VALUES NOTED ON THE DRAWING ARE EFFECTIVE TORQUE ONLY. RUNNING TORQUE IS REQUIRED FOR SELF LOCKING DEVICES AND BOLT FRICTION. RUNNING TORQUE MUST BE MEASURED AND ADDED TO THE EFFECTIVE TORQUE. RECORD RUNNING TORQUE AND FINAL TORQUE ON DRAWING SUPPLIED WITH WORK ORDER.

18. HARNESS ASSEMBLY AND INSPECTION SHALL BE IN ACCORDANCE WITH NASA-STD-8739.4 AND THE FOLLOWING NOTES:

    A. TWIST WIRES BETWEEN 8 AND 16 TIMES THE OUTER DIAMETER OF THE HARNESS.

    B. SEE TENSILE PULL STRENGTHS LIST PER NASA-STD-8739.4.

    C. INSTALL CONTACTS IN ALL UNUSED POSITIONS OF CIRCULAR CONNECTOR.

    D. PERFORM CONTACT RETENTION TEST PER NASA-STD-8739.4.

19. BEFORE STORAGE, PROTECT WITH CONTAMINATION BARRIER. USE CLEAN, NEW, ESD BAG, TOTES OR APPROVED ESD PACKAGING. IDENTIFY PART BY TAGGING WITH PART NUMBER AND SERIAL/LOT NUMBER AS SPECIFIED IN WORK ORDER/SOW PER SAE-AS478, METHOD 35D.

20. REMOVE MANUFACTURER APPLIED LABELS FROM MODIFIED E-FIELD POWER SUPPLIES ITEMS 9 AND 10 PRIOR TO ASSEMBLY

DRAWING LIST

ITEM	QTY	DRAWING NUMBER	DESCRIPTION
1	1	SI101EFA1716	E-FIELD NEGATIVE SUPPLY INTERCONNECT ASSEMBLY
2	1	SI101EFA1719	E-FIELD POSITIVE SUPPLY INTERCONNECT ASSEMBLY
3	1	SI101MFA1709	E-FIELD SUPPLY BASE ASSEMBLY
4	1	SI101MFA1711	E-FIELD SUPPLY PASSIVE CIRCUIT MOUNT ASSEMBLY
5	1	SI101MFD1713	E-FIELD SUPPLY PASSIVE CIRCUIT BASE
6	1	SI101MFD1714	E-FIELD SUPPLY PASSIVE CIRCUIT COVER
7	1	SI101MFD1715	E-FIELD NEGATIVE SUPPLY INTERCONNECT COVER
8	1	SI101MFD1718	E-FIELD POSITIVE SUPPLY INTERCONNECT COVER
9	1	SI101MFP1723	MODIFIED E-FIELD POSITIVE SUPPLY MODULE
10	1	SI101MFP1724	MODIFIED E-FIELD NEGATIVE SUPPLY MODULE

NOTES:

1. CERTIFICATE OF CONFORMANCE REQUIRED FOR ALL COMMERCIAL PARTS.
2. ESD HANDLING AND CONTROL PER ESTABLISHED PROCEDURES.
3. CHROMATE CONVERSION COAT ANY DAMAGED OR MACHINED AREAS ON ALUMINUM PARTS PER MIL-DTL-5541, TYPE I, CLASS 3 (GOLD COLOR).
4. CLEAN ALL SURFACES WITH ISOPROPYL ALCOHOL AND LET AIR DRY. WIPE WITH CLEAN COTTON SWABS AND/OR LINT-FREE CLOTH.
5. HANDLE ALL PARTS AND ASSEMBLIES WITH CLEAN, LINT-FREE, ANTI-STATIC GLOVES OR FINGER COTS.
6. REFERENCE CONNECTION DIAGRAM SI101EFC1721 FOR WIRING.
7. ASSEMBLE AND TEST PER ACME-TST-005.
8. SOLDERING AND INSPECTION SHALL BE IN ACCORDANCE WITH NASA-STD-8739.3.

9. SOLDER ITEM 2 (INTERCONNECT ASSEMBLY) TO ITEM 9 (SUPPLY MODULE) AND ITEM 1 (INTERCONNECT ASSEMBLY) TO ITEM 10 (SUPPLY MODULE). ENCAPSULATE SOLDERED CONNECTIONS WITH ITEM C24 ON SIDE SHOWN.

10. PASS HV TAP WIRE (E15, GRAY) FROM ITEM 2 (INTERCONNECT ASSEMBLY) THROUGH HOLE IN ITEM 4 (PASSIVE CIRCUIT MOUNT) AND ITEM 5 (PASSIVE CIRCUIT BASE). SOLDER TO ITEM C1 (HIGH VOLTAGE DIVIDER) AT LOCATION SHOWN.

11. PASS HV2 RETURN WIRE (E4, BLACK) FROM ITEM1 (INTERCONNECT ASSEMBLY) THROUGH HOLE IN ITEM 4 (PASSIVE CIRCUIT MOUNT) AND ITEM 5 (PASSIVE CIRCUIT BASE). SOLDER TO JUNCTION AT LOCATION SHOWN.

12. PASS WIRES (PIN 16 - 20) OF ITEM 2 (INTERCONNECT ASSEMBLY) THROUGH SLOT IN ITEM 3 (SUPPLY BASE ASSEMBLY). TRIM AND SOLDER TO ITEM 1 (INTERCONNECT ASSEMBLY) PER CONNECTION DIAGRAM.

13. SOLDER ITEM C4 AS HIGH VOLTAGE CONNECTION (RECEPTACLE WIRE-APPROX, LENGTH = 4.67), CRIMP ITEM C28 (ULTRAVOLT WIRE-APPROX. LENGTH = 1.26), AND CRIMP ITEM C25 (22 AWG WIRE-APPROX. LENGTH = 1.23) INTO ITEM C10 (RING TONGUE) 3 PLACES.

14. CRIMP CHASSIS GROUND WIRES FROM ITEM 1 AND ITEM 2 (INTERCONNECT ASSEMBLIES) INTO ITEM C9 (RING TONGUE) 2 PLACES.

15. STAKING, BONDING AND INSPECTION SHALL BE IN ACCORDANCE WITH NASA-STD-8739.1.

THIS DRAWING IS UNDER CONFIGURATION CONTROL: DO NOT REVISE WITHOUT APPROVED ECO

APPROVED: E.McCLOSKEY DATE: 1/6/15

CHG	ZONE	DESCRIPTION	APP/DATE
B		INCORPORATED ACME-ECO-397	KR 12/30/14 AD 12/31/14 GF 01/05/15
A		INCORPORATED ACME-ECO-039	AD 08/28/12 MP 08/28/12 GF 08/24/12

REVISIONS

UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES

TOLERANCE ON			
.X	± 0.03	PROJECT LEAD	B. BOROWSKI 01/13/12
.XX	± 0.01	QUALITY	D. SEELEY 01/13/12
.XXX	± 0.005	SAFETY	W. BORRELLI 01/12/12
FRACTION	± -	ENGINEER	A. DRAKE 01/11/12
ANGULAR	± 0.50°	DESIGNER	G. FEDEVICH 01/11/12
BREAK EDGES	N/A	REVIEWER	M. PREVITE 01/11/12

DIMENSIONING AND TOLERANCING PER ASME Y14.5M-1994

PRIOR TO USE CHECK FOR ANY OUTSTANDING ECO'S WITH CONFIGURATION MANAGEMENT OFFICE

ZIN TECHNOLOGIES
6745 ENGLE ROAD
AIRPORT EXECUTIVE PARK
CLEVELAND, OHIO 44130
440 625-2200

CONTRACT NUMBER: NNC09BA02B
DELIVERY ORDER: SPACE 110
MODEL NAME: SI101MFA1708

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION
JOHN H. GLENN RESEARCH CENTER
LEWIS FIELD
CLEVELAND, OHIO

ACME CHAMBER INSERT ASSEMBLY, LAUNCH CONFIGURATION

SI101MFA0100

E-FIELD POWER SUPPLY ASSEMBLY

SIZE	DRAWING NUMBER	REV	BLDG	AREA
D	SI101MFA1708	B	ZIN	GRC

UNITS: INCH

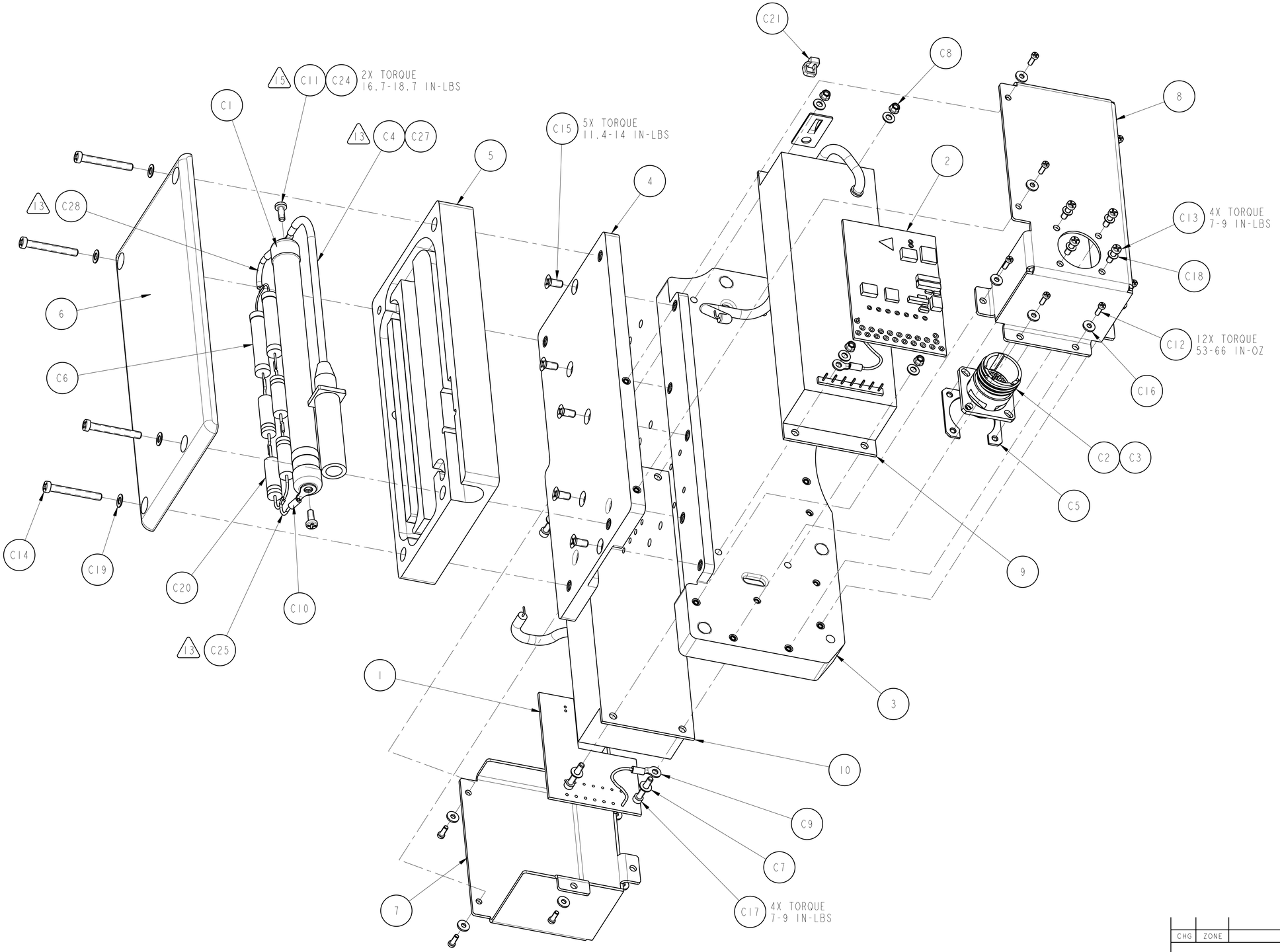
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1/1	01/13/2012	1 OF 3	MECHANICAL

PRO/E VERSION: Wildfire 2.0

REV B

SIZE D DRAWING NUMBER SI101MFA1708

FOR NOTES SEE SHEET 1



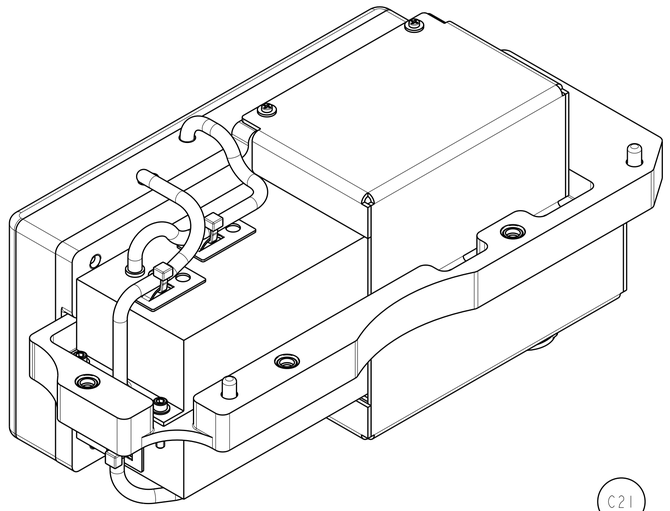
CHG	ZONE	DESCRIPTION	APP/DATE
REVISIONS			

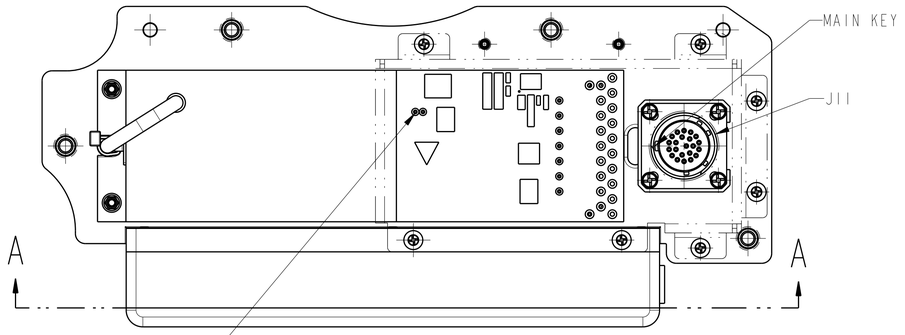
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D	S1101MFA1708	B	ZIN	GRC
			UNITS: INCH	
SCALE: 1/1	INITIAL REL DATE 01/13/2012	SHEET 2 OF 3	DISCIPLINE MECHANICAL	

PRO/E VERSION: Wildfire 2.0

REV B
SIZE D DRAWING NUMBER S1101MFA1708

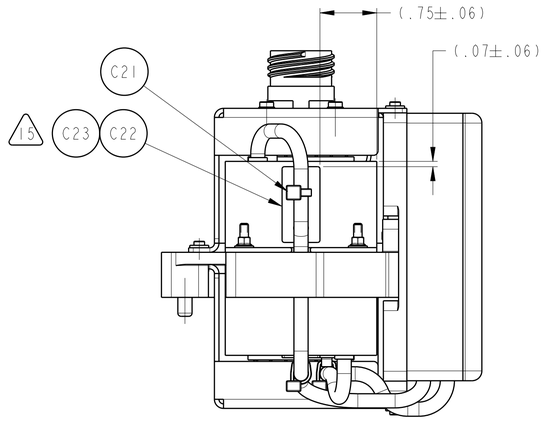
FOR NOTES SEE SHEET 1



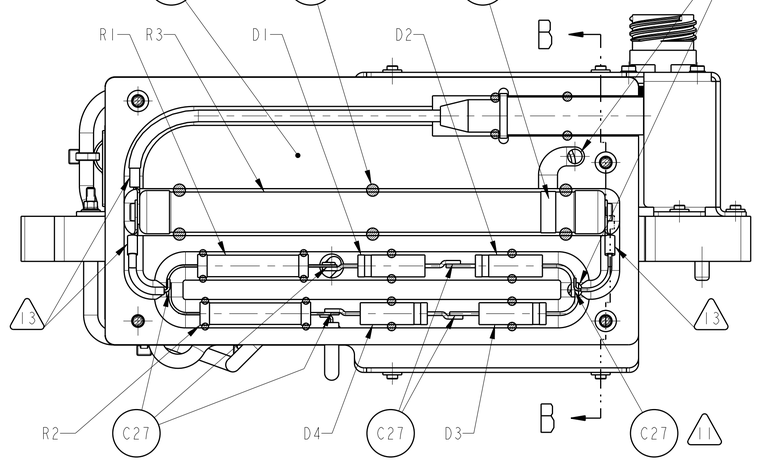


9 C24 C27 (9 PIN LOCATIONS)

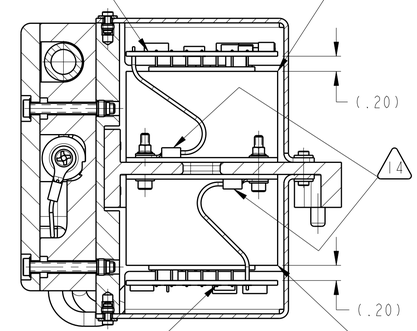
15 C24 (WIRES NOT SHOWN - SEAL HOLES)



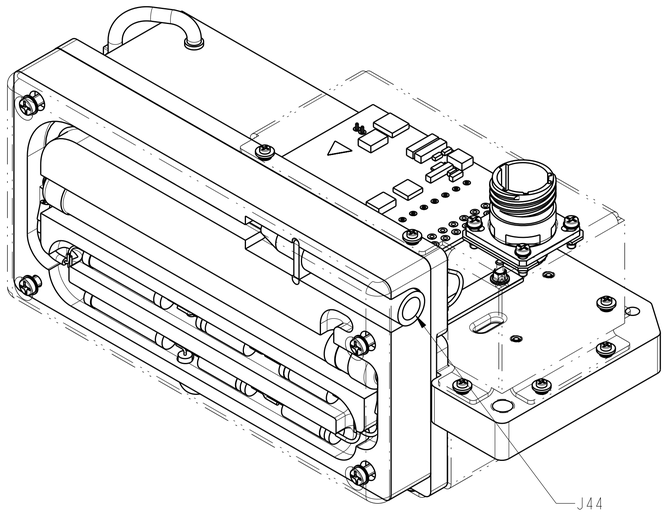
16 C26 15 C24 TYP. 10 C27 (WIRE NOT SHOWN)

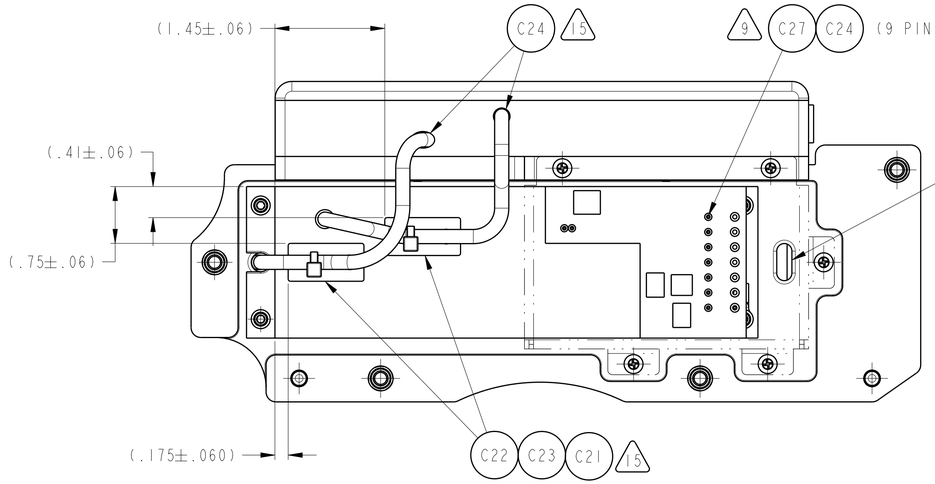


SECTION A-A



SECTION B-B





CHG	ZONE	DESCRIPTION	APP/DATE
REVISIONS			

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			UNITS: INCH	
SCALE: 1/1	INITIAL REL DATE 01/13/2012	SHEET 3 OF 3	DISCIPLINE MECHANICAL	

PRO/E VERSION: Wildfire 2.0

SIZE D DRAWING NUMBER S1101MFAI708 REV B